

CSE 332: Computer Architecture And Organization

Assignment#1

Spring 2025

Name: _____

Student ID: _____

Direction: You can consult any resources such as books, online references, and videos for this assignment, however, you have to properly cite and paraphrase your answers when it is necessary. There will be points for partial attempts. You can upload a typed or handwritten copy of your assignment to the canvas. However, you have to show the original copy of your assignment in case of arising questions about the authenticity. You should upload your assignment to canvas by the due date on canvas.

Submission guidelines: All our assignments will be based on 100 points so that we can assign weights to get your points for final grade. Whoever fails to submit within the time period assigned through NSU canvas but submit the assignment within the next three days will be punished by 20% deduction of points (e.g. start with -20 points within 100 points). The late submission will be open until semester's end through canvas. After the first 2 days of grace period, the deduction will be 40%.

Note: You are allowed to use only instructions implemented by the actual MIPS hardware provided in the Canvas references. Use assembly language format from the references or the book. Note, base ten numbers are listed as normal (e.g. 23), binary numbers are prefixed with 0b or format such as XX_2 and hexadecimal numbers are prefixed with 0x or format such as XX_{16} / XX_{HEX}

Problem (50x2=100 points): Write definition, elaboration, figure, flowchart or any suitable method to describe the following Computer Architectural and Digital Design Concepts.

1. BJT, MOSFET
2. Diode and LED (Light Emitting Diode)
3. pMOS, nMOS, CMOS
4. Universal Logic Gate
5. CMOS 3 input NAND Logic
6. Process Node Technology
7. Edge Triggered D Flip-Flop
8. Clock Edge
9. Combinational Logic
10. Sequential Logic
11. Moore State Machine

12. Register Transfer Level (RTL)
13. Timing Diagram
14. Karnaugh Map
15. Multiplexer (MUX)
16. Demultiplexer (DEMUX)
17. Decoder
18. Encoder
19. Full bit Adder
20. Ripple Carry Adder
21. Comparator
22. Counter
23. Shift Register
24. Ring Oscillator
25. Clock Divider
26. Asynchronous Design
27. Setup Time
28. Hold Time
29. Propagation Delay
30. Fan-out
31. Photolithography in chip fabrication
32. RTL Synthesis
33. HDL (Hardware Description Language)
34. EDA (Electronic Design Automation) example
35. VHDL, Verilog
36. FPGA (Field-Programmable Gate Array)
37. CPLD (Complex Programmable Logic Device)
38. ASIC (Application-Specific Integrated Circuit)
39. Gate-Level Design
40. Testbench
41. DFT (Design for Testability)
42. Logic Synthesis
43. Power Optimization
44. Heterogeneous Multicore Processor
45. Cache
46. Main Memory
47. Name five steps between sand to chip
48. Name five stages between HDL to Chip layout
49. RISC, CISC
50. Wafer, Die, Chip Packaging